Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 21)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Describe the main principles of the circular economy.

[6 marks]

0 1 . 2 Explain what is meant by 'optimum production rather than maximum production', using a comparison between a natural ecosystem and an agroecosystem.

[2 marks]

0 1 . 3 Explain why international cooperation (eg through the EU or UN) is needed to implement circular economy initiatives.

[2 marks]

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0 2 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

0 2 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

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0 3 . 1 Describe how each of the following threatens Antarctica.

[4 marks]

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Global climate change .....

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Ozone depletion .....

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Tourism .....

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Future mineral exploitation .....

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0 3 . 2 Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

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0 4 Table 1 shows the estimated population of blue whales and the number killed by whaling in selected years.

Table 1

Year	1900	1930	1960	1986	2020
Estimated population	240 000	100 000	5 000	2 000	15 000
Whales killed that year	200	30 000	1 200	0	0

0 4 . 1 Calculate the percentage decline in the blue whale population between 1900 and 1986.

[1 mark]

Answer ☐ %

0 4 . 2 Explain why whale populations were so easily over-exploited.

[3 marks]

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0 4 . 3 Describe how the International Whaling Commission regulates whaling.

[4 marks]

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0 4 . 4 Explain why the ITTO has been less successful than the IWC in achieving sustainable exploitation.

[2 marks]

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0 5 . 1 Explain **three** features of natural systems that make them sustainable.

[3 marks]

0 5 . 2 Explain why the human use of mineral resources makes their sustainable exploitation much harder than the use of materials in a natural cycle.

[2 marks]

0 5 . 3 Explain **three** principles of the circular economy that should be applied when a new product is designed.

[3 marks]

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0 5 . 4 Explain why an activity that is sustainable for a small population may be unsustainable for a large one. Use **one** example.

[3 marks]

0 5 . 5 Explain why urban populations are less likely than rural communities to notice that a resource is being over-exploited, and give **two** consequences of this.

[4 marks]

15

0 6 . 1 Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

0 6 . 2 Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

0 6 . 3 Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

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0 7 Table 1 shows the number of plant species per m^2 in a chalk grassland reserve under different management, ten years after management began.

Table 1

Management	Mean plant species per m^2	Percentage of area covered by scrub / bushes
No management (abandoned)	8	55
Sheep grazing	24	5
Annual mowing (August)	19	0
Winter burning	14	10

0 7 . 1 Explain what is meant by a **plagioclimax** community and why chalk grassland is an example.

[2 marks]

0 7 . 2 Use Table 1 to explain why grazing produced the highest plant diversity.

[4 marks]

0 7 . 3 Explain why different grazing animals are chosen for different plagioclimax habitats.

[2 marks]

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0 7 . 4 Explain how burning maintains heathland.

[2 marks]

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0 8 A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 8 . 1 Use the formula $D = N(N - 1) / \Sigma n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

Answer

0 8 . 2 Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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0 9 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 9 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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1 0 Table 1 shows some medicines that were discovered in wild species.

Table 1

Medicine	Source species	Use
Taxol	Yew tree	Cancers of the breast, ovary, lung, bladder and prostate
Quinine	Cinchona tree	Malaria
AZT	Caribbean marine sponge	HIV / AIDS
Diosgenin (steroids)	Mexican yam	Contraceptive pill, cortisone

1 0 . 1 Explain why many plants produce chemicals that turn out to be useful medicines.

[2 marks]

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1 0 . 2 Explain why the medicines in Table 1 provide an argument for conserving whole habitats rather than individual species.

[2 marks]

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1 0 . 3 Describe how **one** named species is being studied to help solve a human health problem other than by providing a medicine.

[2 marks]

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1 0 . 4 Discuss the extent to which the resources and ecosystem services provided by other species justify the conservation of biodiversity.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

OR

1 1 . 2 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS